

Protecting the Past, Securing the Future: The Twin Pillars of Insurance in Disaster Management

Summary

In recent years, the frequency of extreme weather events has increased the amount of insurance claims, and it has become increasingly difficult, if not impossible, to obtain coverage in certain areas. The insurance industry has to deal with the increasing frequency of disasters as well as ensure its own risk resilience. So should insurance companies continue to underwrite in today's world of extreme weather? Do real estate developers continue to build? And how are buildings at risk protected and valued? The need to address these questions is urgent.

To address the first question, we develop an **ARIMA loss prediction model** and an insurance model to quantitatively determine insurance companies' underwriting decisions. We use the future economic losses predicted by the ARIMA model as the cost, and bring the cost into the **insurance model** to calculate the premiums and the number of enrollees when the insurance company reaches the break-even point, and then compare the data with the historical data to decide whether or not to continue to underwrite the insurance in that place, and then finally, we bring in the predicted data from Japan and the U.S.A. and conclude that it is better to continue to underwrite the insurance in Japan rather than the U.S.A. We conclude that we should continue to underwrite in Japan and not the U.S.

For question two, we used a **hierarchical cluster analysis** and a modified insurance model to evaluate whether to proceed with construction. Hierarchical cluster analysis was performed by the number of tornado, hail and thunderstorm wind occurrences, and the results were categorized into high-risk, medium-risk and low-risk areas. Then the principle of **price discrimination** was combined to improve the insurance model and finally concluded that construction is not recommended in high-risk areas, if construction is carried out in medium-risk areas, it is recommended to increase the premium by 28.57%, while for low-risk areas it is recommended to carry out construction and reduce the premium by 14.29%.

For question three, because there are many qualitative factors to be considered when assessing the value of buildings and the level is complex, the **AHP model** is used to combine multiple factors to assess the buildings. Firstly, five indicators such as historical value, cultural significance and economic contribution are selected as the evaluation layer, secondly, three specific famous buildings are selected as the program layer, and finally, the judgment matrix of each layer is constructed according to the relevant literature and consistency test is carried out to get the weights of each layer and the total weights. This paper selects the Statue of Liberty, Neal Hall and Fortress Monroe as the program layer and evaluates its value, and the results show that the Statue of Liberty has the highest weight of 0.49, so the Statue of Liberty has the highest value and should be emphasized on its preservation and protection. Finally, based on these conclusions, a letter of appeal for the protection of the Statue of Liberty was written from the aspects of cultural significance, social significance and restoration measures of the building.

Keywords: ARIMA; Hierarchical Cluster Analysis; Insurance Model; AHP

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1 Introduction

1.1 Problem Background

In recent years, extreme weather events such as extreme cold, extreme drought, high temperatures, blizzards and torrential rains have occurred frequently.

The insurance industry has to enhance the coverage rate of catastrophe insurance, but also to face the increasingly frequent disasters brought about by the increase in the amount of compensation, but also to ensure their own risk resilience. How can the insurance industry diversify its own risks in the face of frequent occurrence of extreme weather?

Extreme weather events are becoming a crisis for property owners and insurers. Insurers are being forced to increase insurance costs in order to survive, and climate change is expected to increase insurance costs by 30-60% by 2024. In addition, the global average insurance coverage gap worldwide is 57% and growing. This highlights the plight of the insurance industry - there is a crisis in the profitability of insurance companies and the affordability of property owners. Therefore, it is urgent to study the sustainability of property insurance.



Figure 1: The Hazards of Extreme Weather

1.2 Restatement of the Problem

Considering the background information and constraints identified in the problem statement, we need to address the following questions:

- As the frequency of extreme weather events grows globally, the amount of compensation paid for them by the property insurance industry is also rising, and under these conditions, insurance companies need to fully consider all aspects of underwriters and set different premiums in order to maximize their returns.
- The insurance landscape is constantly changing due to extreme weather. In the future, communities and real estate developers will need to consider how to make real estate decisions in light of the changing insurance landscape, and they should evaluate the feasibility of building on specific sites. This will ensure that properties are more resilient and able to adapt to future challenges.
- Define the indicators for evaluating the conservation value of a building and use a multi-indicator decision analysis method to synthesize the conservation value of a building and its weights, and then determine the degree of conservation measures

that need to be taken based on the building's conservation value score.

- We selected a historic building vulnerable to extreme weather, valued the landmark in conjunction with a value assessment model, and finally integrated the results of the analysis and preservation plan into a letter of recommendation to the community emphasizing the importance and urgency of preserving the landmark..

1.3 Our Work

Task 1: We will build an ARIMA model to predict the amount of claims for two regions with extreme weather events located on different continents.

Task 2: We extend our insurance modeling to assess the feasibility of building on a given site. This means that we not only consider the impact of extreme weather events on existing properties, but also how these risks can be avoided in future builds.

Task 3: Considering that many culturally or socially significant properties are too risky for insurance companies to insure, we developed a heritage value model that evaluates multiple dimensions of historical value, cultural significance, and economic contribution so that community leaders can better identify these buildings and quantify the extent to which preservation measures should be taken.

Task 4: Prepare a letter with recommendations to the community. In the letter, explain the results of the insurance model and the preservation model, emphasizing the significance of the landmark and the risks it faces. Based on the results of the modeling, propose specific preservation measures, including rehabilitation, structural reinforcement, and construction of disaster prevention facilities. Emphasize the urgency and importance of protection and encourage the community to work together to protect their cherished landmark.

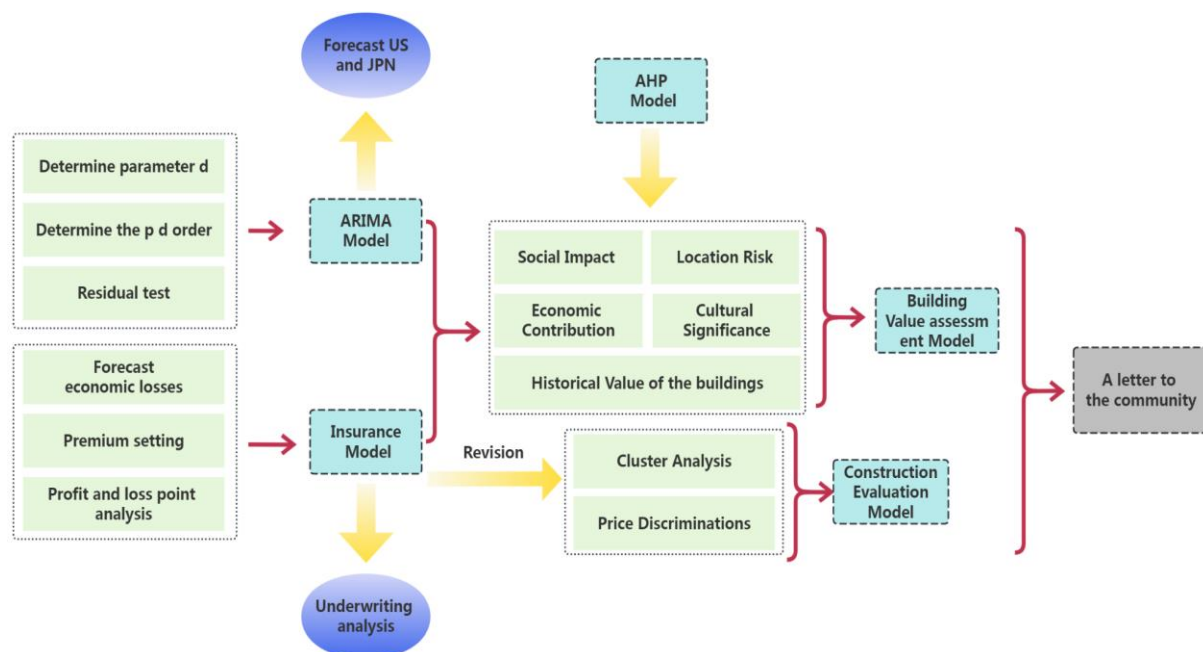


Figure 2: Our Work

2 Assumptions and Justifications

Assumption 1: In this paper, the frequency of extreme weather in the United States represents the frequency of extreme weather in North America, and the frequency of extreme weather in Japan represents the frequency of extreme weather in Asia.

Justification: The United States spans multiple climate zones, which to a certain extent can reflect the overall climate dynamics of North America; Japan is located in the Pacific Rim seismic zone, which is the main passing area of typhoons in the western Pacific Ocean, and to a large extent represents the extreme weather trends in the whole of Asia.

Assumption 2: There is some similarity and diversity among the major U.S. states, and extreme weather events are primarily caused by local climate change anomalies.

Justification: Reasonableness of assumptions: There are similarities between geographic locations in close proximity to each other, and these geographic similarities may lead to certain commonalities in the natural environments, climatic characteristics, vegetation types, etc. of these states, which may make them show similar trends in terms of the frequency of extreme events. Moreover, the United States is a vast country with geographic locations covering multiple lines of latitude and longitude, and the diverse climate types make for a variety of clustering categories across continents.

Assumption 3: Decision makers can make pairwise comparisons of elements at each decision level using a scale of 1 to 9, where 1 indicates equal importance or priority and 9 indicates extreme importance or priority.

Justification: Can help decision makers break down complex problems into simpler, more manageable components and organize them in a clear, coherent manner. Facilitates the calculation of the relative importance or preference of each element and its aggregation into a final ranking or score of alternatives.

Assumption 4: It is acceptable for decision makers to use the eigenvector method to derive the weights or priorities of elements.

Justification: The method is based on the principle of maximum eigenvalue and the Perron-Frobenius theorem. It expresses the decision maker's subjective judgments and preferences in a quantifiable and comparable manner and incorporates both tangible and intangible criteria.

Assumption 5: There is no high correlation between the indicators used in this paper.

Justification: For example, using the number of floods, tornadoes and storms as indicators for cluster analysis, the occurrence of each extreme weather event is largely dependent on the specific geographic and climatic conditions of the area, and there is no generalized, high statistical correlation between them.

3 Notations and Definitions

3.1 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations

Symbol	Description
p	Autoregressive Order
d	Difference Order
q	Moving average Order
D_{ij}	Distance between Nearest Samples
W	Weight Vector
λ	Eigenvalues
CI	Consistency Indicator
RI	Consistency Ratio
CR	Consistency Ratio
P	Corporate Net Profit
I	Insurance Premium Per Person
k	Corporate Fixed Costs
N	Number of Participants
L	Economic Loss
C	Total Costs
F	Basic Insurance Premium Per Person
E	Expected Profits
E/N	Apportionment of Expected Profits
L/N	Apportionment of Economic Loss
$w^{(2)}$	Criteria Layer Weights
$w^{(3)}$	Alternative Layer Weights

3.2 Definitions

Insurance Protection Gap: The difference in protection coverage between economic losses brought about by natural disasters and the amount of those losses that are covered.

Underwrite: Accept liability for, thereby guaranteeing payment in the case of loss or damage.

Price Discrimination: Essentially a price differential, which usually means that the provider of a good or service, when offering the same grade of a good or service of the same quality to different recipients, applies a different selling price or rate between the recipients.

4 Model Establishment of Problem 1

4.1 Loss Prediction Model

Autoregressive Integrated Moving Average Model (ARIMA) is a model that analyzes the dynamics of time series variables based on AR and MA models, and because of its accurate forecasting accuracy, ARIMA has become one of the most commonly used linear models in

time series forecasting. ARIMA has combines three basic methods:

Autoregressive model: predicts the future performance of the same variable by its historical performance and assumes that they are linearly related. The AR model order recorded as a p-value, which is calculated as follows:

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \dots + \alpha_p y_{t-p} + \varepsilon_t \quad (1)$$

Integration: the number of differences made for the time series data to become smooth, denoted by the d-value for order.

Moving average model: focuses on the accumulation of the error term in the autoregressive model, with the q-value representing the model order, which is calculated as follows:

$$y_t = \varepsilon_t + v_1 \varepsilon_{t-1} + v_2 \varepsilon_{t-2} + \dots + v_q \varepsilon_{t-q} + c \quad (2)$$

Step 1: Stabilization

The premise assumption used in the ARIMA model is that the time series data must be smooth, for this we use the unit root test to test the smoothness of the series between the two countries of the U.S.A. and Japan, and if the test results are unstable we will use the difference method on the data to convert the non-smooth process to a smooth process.

Table 2: P Value of US and JPN Time Series Before and After Difference

	P value before difference	P value after difference
US	0.0653	0.0000
JPN	0.0000	—

Step 2: Determination of Order

Bayesian Information Criterion (BIC) is an information criterion function that can be used to determine the optimal order of a model, which is constructed based on the likelihood function. We will calculate the BIC values of the model at different orders based on historical data by computer programming cycles to find the p and q that minimize the BIC, i.e., the optimal order of the model.

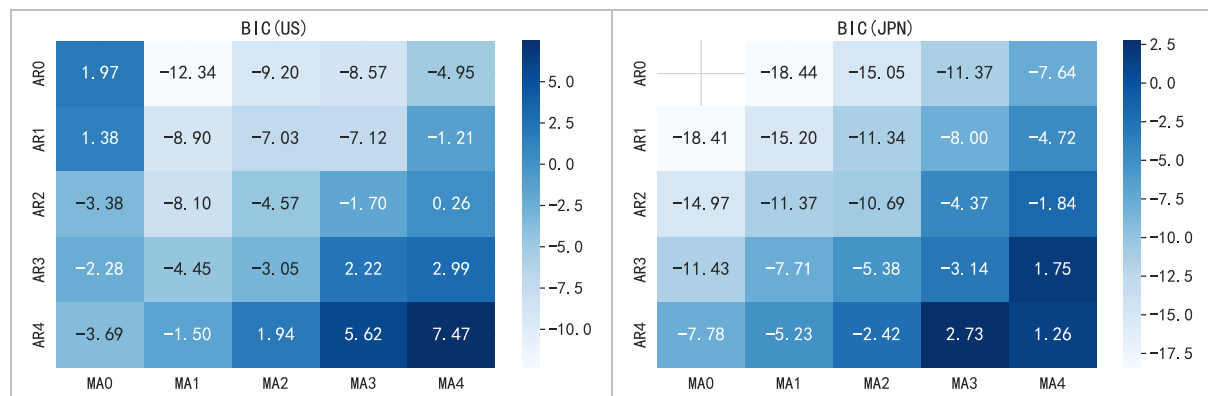


Figure 3: BIC Values of US(Left) BIC Values of JPN(Right)

From the above figure, it can be seen that the p, q order corresponding to the lowest value of BIC in the U.S. and Japan are both 0,1. Therefore, the ARIMA model for the U.S. is ARIMA(0,1,1), and the ARIMA model for Japan is ARIMA(0,0,1).

Step 3: Residual Test

According to the ARIMA model established above, we will carry out the residual autocorrelation coefficient test to verify the stochasticity of the residual series, and the test results found that the lagged order of the autocorrelation coefficient basically falls within the confidence interval, which indicates that the fitted data are independent and can be predicted.

Step 4 Prediction

We project the economic losses that will occur in both countries over the next decade based on past economic losses caused by extreme weather events in both countries.

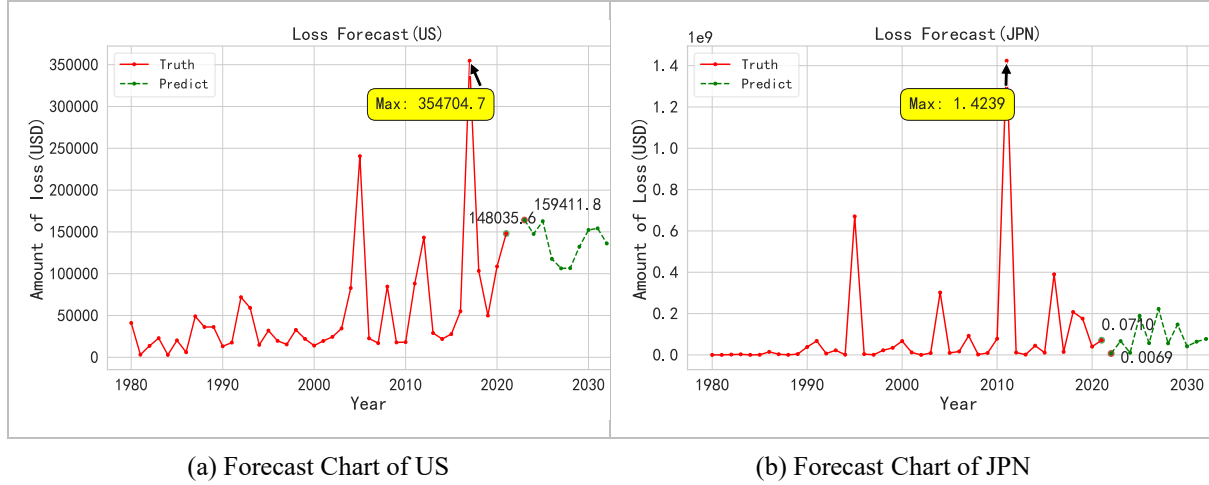


Figure 4: Forecast Chart in One Cycle

As can be seen from the above chart, the trend of economic losses in the United States and Japan in the next 10 years is cyclical, but basically maintained between the average economic losses in recent years, that is, based on the data in recent years in the future foreseeable economic losses, the insurance company must be for different regions to establish a wide range of insurance models applicable.

4.2 Insurance Model

In this paper, we will take the year 2030 as an example, and calculate the premiums collected by insurance companies in that year through economic loss projections, and subject them to a break-even point analysis to arrive at a minimum number of participants in different regions. Therefore, we define the relevant variable parameters to quantify the insurance model.

The modeling is as follows:

$$\begin{cases} P = I \times N \\ I = F + \frac{E}{N} + \frac{L}{N} \\ C = k + L \end{cases} \quad (3)$$

In the above formula, the total participation premium per person (I) constructed according to different regions consists of the base premium rate (F), the equalized portion of the expected return (E/N), and the equalized portion of the projected economic loss in 2030 (L/N), while the total cost (C) consists of the fixed cost and the economic loss (L), and we can calculate the expected return of an insurance company in applying the insurance model based on the spread of the total profit (P), which consists of the number of participants who pay the premiums (I×N),

and the total cost (C), we can calculate the expected return of the insurance company in using the insurance model.

$$BEP(C) = I \times N_{min} \quad (4)$$

Based on the results of the above equations, we construct the critical number of people that exactly offsets the cost of expenses in the profit and loss analysis, i.e., the minimum number of local enrollees (N_{min}).

In the North American insurance model, fixed cost (k) is \$418 million for plant and equipment in Allstate's financial statements, basic enrollment cost (F) is the average U.S. property and casualty insurance enrollment premium of \$1,400 per year, expected earnings (E) is \$100 million of net income in the company's 2022 financial statements, and enrollment (N) is the most recent Affordable Care Act enrollment of 20 million.

In the Asian insurance model, fixed costs (k) is \$174 million for plant and equipment in the financial statements of Japan Life Insurance Company, basic enrollment costs (F) is the average enrollment fee of \$1,250 per year for property insurance in Japan, expected earnings (E) is net income of \$64 million in the company's 2022 financial statements, and enrollment (N) is the most recent enrollment figures released by the Ministry of Health, Labor and Welfare of Japan 45.13 million people. By solving the above equation, it can be seen that for the North American insurance company to obtain the expected revenue, it needs to price property insurance at \$1,479.65/year and the insurance company has 1,291,522 underwriters in order to reach the break-even point, while for the Asian insurance company to obtain the expected revenue, it needs to price property insurance at \$1,300.14/year and 1,787,431 underwriters in order to reach the break-even point. The comparison of historical and projected data shows that the U.S. region is not suitable for continued underwriting while the Japanese region can continue to be underwritten. In summary, insurance companies need to set different participation standards for different continents in order to maximize their own benefits and underwriting decisions under the same risk.

5 Model Establishment of Problem 2

5.1 Modeling Ideas

In order to assess where and whether to build in certain locations, this paper analyzes the data by collecting relevant data about the impacts of extreme weather across the United States.

First, the data is processed to obtain data such as the number of extreme weather occurrences and damages caused by each state in the U.S., and heat maps are used to visualize the differences in risk. Second, the number of tornadoes, the number of hailstorms, and the number of thunderstorms in the U.S. states are systematically clustered by three quantitative indicators, and the U.S. states are classified into high-risk, medium-risk, and low-risk areas by the clustering results. Finally, by improving the insurance model and borrowing the principle of price discrimination, different insurance rates and premiums are charged to states in different risk regions of the United States. That is to say, by comparing the high and low risks and the high and low premiums to better quantitatively describe the high and low benefits and costs of building in a certain place.

5.2 Geospatial Risk Analysis Model: Based on HCA Model

Step 1: Data Collection and Analysis

Gather geographic, climatic, and historical data on extreme weather events in different regions across the U.S. Perform a preliminary analysis of the data. Calculate the number, frequency, and geographic location of extreme weather events in each U.S. state. Obtain a risk assessment for each state.

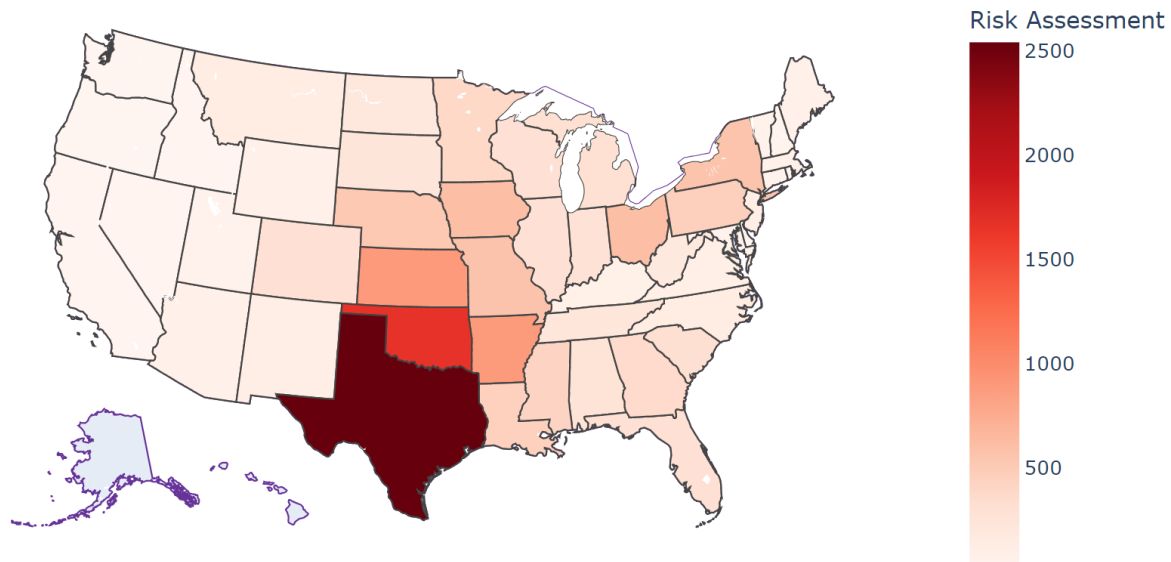


Figure 5: Risk Assessment(US)

Step 2: Hierarchical Cluster Analysis (HCA)

The use of hierarchical Cluster analysis utilizes the significance of the results to divide regions and select different insurance models in different regions.

Systematic clustering is the most used method in cluster analysis. The basic idea is that the nearest variables are clustered into one class first, and the distant ones are clustered into classes later, and this process keeps cycling so that each sample is always clustered into the appropriate class. Systematic clustering is also known as hierarchical clustering, because the results will show a significant hierarchical structure, usually divided into bottom-up and top-down, the principle of the two methods is the same, the difference is that the direction of the calculation is opposite.

Bottom-down: also known as the merger method, this method is first of all, each sample as a separate class, and then through the distance calculation, will be close to the two samples merged into a class, the other samples are still each for a class, repeat the above process until the set number of clusters. Top-down: also known as decomposition method, and since the opposite of low down, will see all the samples to see a class, through the distance between the samples, selected the distance between the farthest two samples, each for a class, the rest of the samples according to the principle of proximity to the allocation of the two categories, repeat the above process until the number of clusters to reach the set. Here we use the merger method used.

After we know the method of clustering, the next key issue is to define the distance. In the clustering of common distance definition methods are absolute value distance, Euclidean distance, Minkoff distance, Chebyshev distance, etc., the most commonly used is the Euclidean

distance, so we use the Euclidean distance, that is, Euclidian distance.

$$d_{ij} = \sqrt{\sum_{k=1}^m (x_{ik} - x_{jk})^2} \quad (5)$$

After the problem of distance definition is solved, it is the turn of how to determine what is the distance between two classes, and the usual methods of determination are the shortest distance method, the longest distance method, the intermediate distance method, the class averaging method, the center of gravity method, and the method of deviation leveling. In this paper, we use the shortest distance method.

The shortest clustering method defines the distance between classes G_i and G_j as the distance between two neighboring samples between the two classes, i.e., the distance D_{ij} between classes G_i and G_j is defined as:

$$D_{ij} = \min_{\omega_s \in G_i, \omega_t \in G_j} d(\omega_s, \omega_t) \quad (6)$$

We used the number of tornadoes, storms, and floods in each U.S. state in 2023 as metrics for regionalization and selection of different insurance models. First, we need to standardize the three metrics to ensure that each metric has equal weight in the clustering process. Normalization usually involves converting the data to a form with a mean of 0 and a standard deviation of 1.

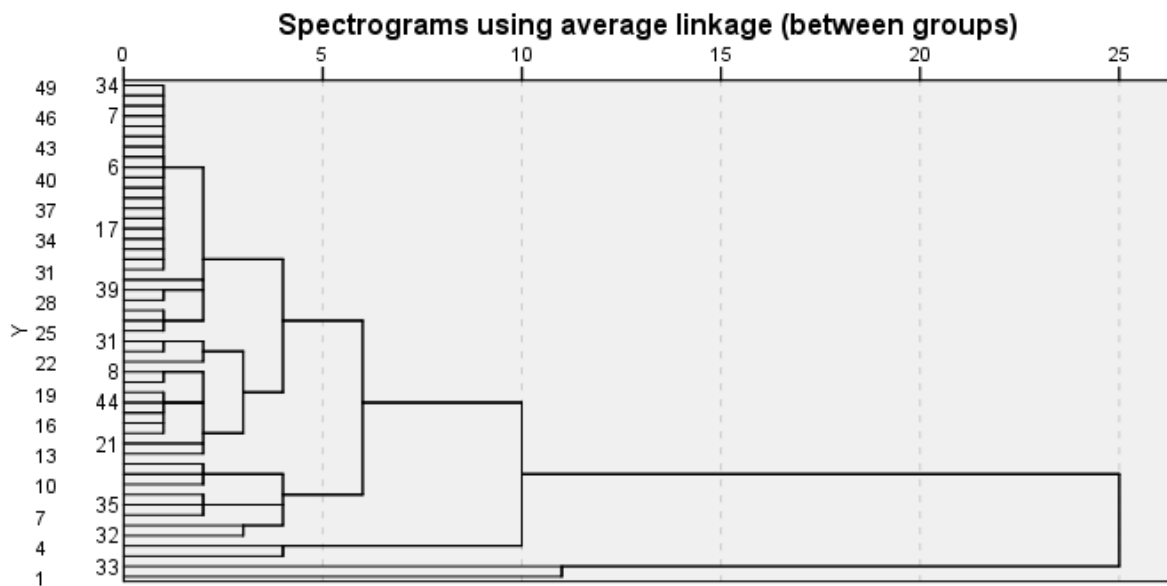


Figure 6: Clustering Result

By calculating the Euclidean distance between all states, we can construct a distance matrix. We then use a systematic clustering algorithm to group states based on this distance matrix. The systematic clustering algorithm starts with each state as a separate cluster and then gradually merges the closest clusters until all states are in the same cluster.

During the clustering process, we can choose a threshold to determine when to stop merging clusters. Finally, based on the clustering results, we are able to develop appropriate insurance models and investment strategies based on the level of extreme weather risk in different states.

The frequency of extreme events in the United States by continent is shown in the chart

above, and it is clear that the frequency of extreme events in most areas of the United States is in the range of 1,000 to 1,500, and most of the zones are located in the latitude of 30 degrees to 45 degrees north, and the longitude of 80 degrees to 110 degrees east. Combined with the range of latitudes and longitudes across the U.S. and the local climate, real estate developers should try to invest in locations with latitudes and longitudes ranging from 24 degrees to 30 degrees north and 110 degrees to 125 degrees east.

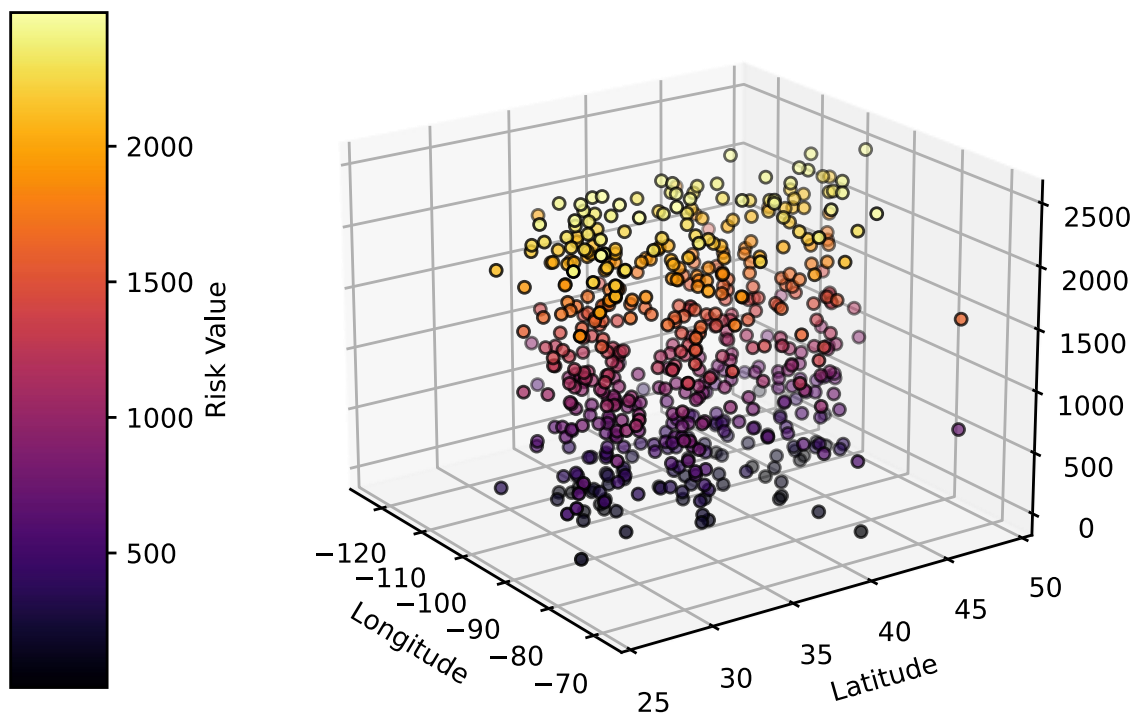


Figure 7: Risk Location Map

By categorizing the risk indicators for each continent in the U.S. and combining the principle of price discrimination, this paper adjusts the above insurance model as follows:

For regions with fewer extreme weather events, i.e., those states categorized as low-risk in the cluster analysis, local residents will have low willingness to underwrite insurance due to the lower risk they bear, and insurance companies appropriately reduce the base premium from the original 1,400 U.S. dollars per year to 1,200 U.S. dollars per year, in order to attract local enrollment with price concessions.

For areas with moderate extreme weather events, i.e., states that are categorized as medium risk in the cluster analysis, local residents will have different attitudes toward coverage due to different levels of risk appetite, so the insurance model can be fine-tuned to implement different base premium prices for different enrollees' needs using the principle of first-degree price discrimination.

For areas with a high incidence of extreme weather events, i.e., states categorized as high risk in the cluster analysis, where residents may choose to enroll due to higher risk taking, insurers could increase the base premium to \$1,800 per year in the model, and exclude or shorten the duration of guarantees for areas expected to be very high risk.

6 Building Value Assessment Model: AHP Model

6.1 Modeling Ideas

In order to develop a preservation model for identifying buildings in a community that should be protected and preserved, we need to consider several factors, including cultural, historical, economic, and social impact. The model will help community leaders assess the preservation value of each building and determine the extent of preservation measures to be taken accordingly.

First, start with data collection. Detailed information about each building in the community is collected, including architectural history, architectural function, economic contribution, and social engagement. Second, a multifaceted overview and summary of the value of the buildings is conducted based on the acquired data to be used in the definition of evaluation indicators, such as the historical value of the buildings (HV), cultural significance (CS), economic contribution (EC), social impact (SI), and location risk (LR).

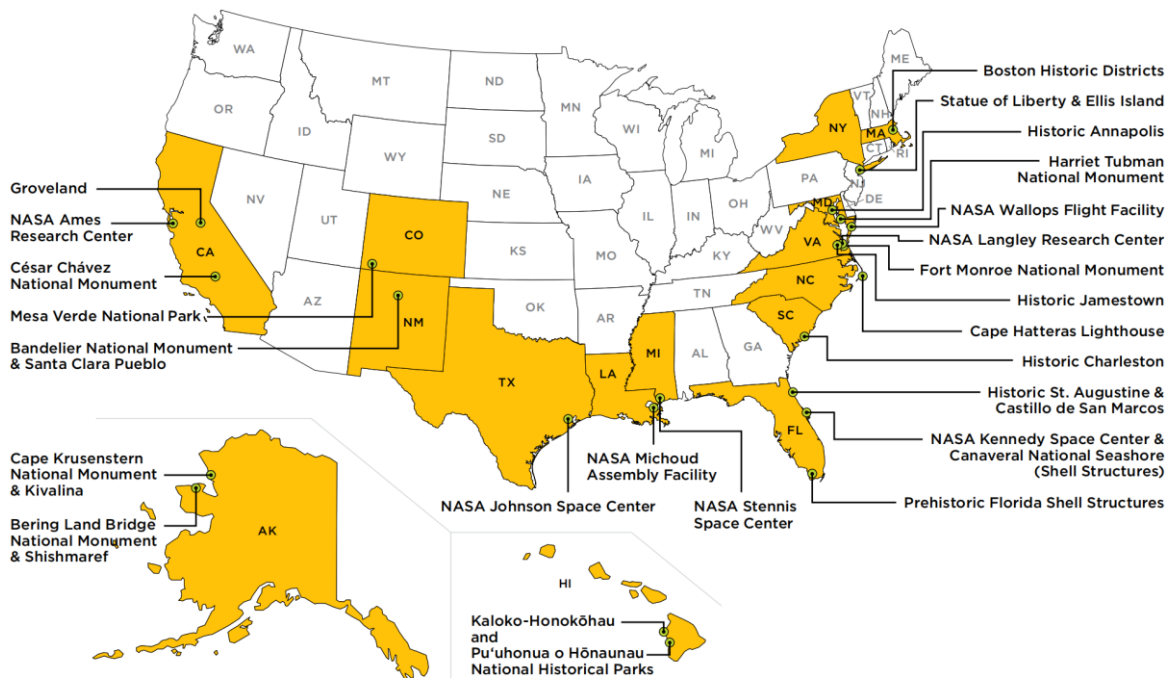


Figure 8: Distribution of Famous Buildings(US)

Finally, the weight of each evaluation indicator was determined by expert evaluation or community survey, and the evaluation indicator system was used as the criterion layer of the AHP model to comprehensively evaluate the conservation value of the buildings.

After calculating the comprehensive conservation value score for each building it is possible to determine the extent of conservation measures that need to be taken based on its score and community resource constraints.

6.2 AHP Model and Evaluation Indicators

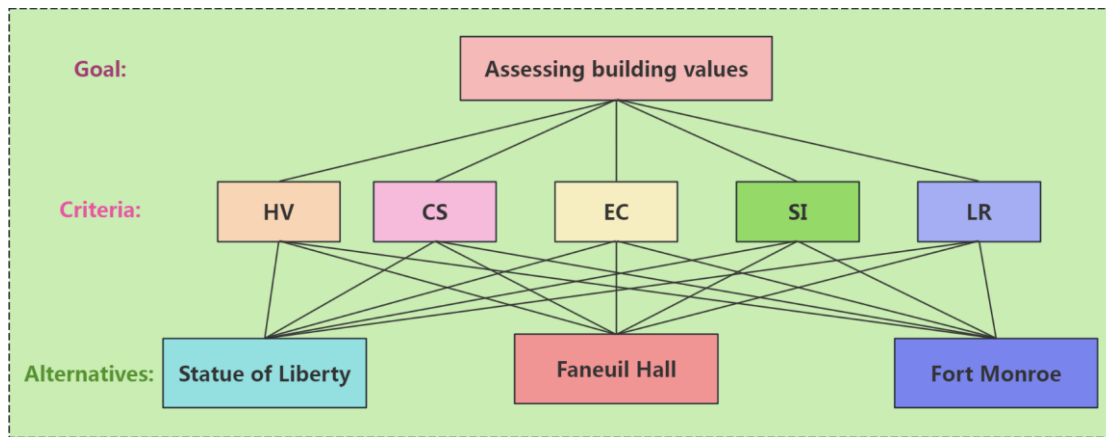


Figure 9: AHP Model

A multi-indicator decision analysis was constructed and analyzed using the AHP model. Where the objective layer of the AHP model is to assess the value of the building; the evaluation layers are historical value (HV), cultural significance (CS), economic contribution (EC), social impact (SI), and location risk (LR); and the decision-making or selection layers are the three famous historical buildings, which are the Statue of Liberty, Faneuil Hall, and Fort Monroe.

6.3 AHP Model Solution

In this paper, the main purpose is to establish the AHP evaluation model through Python, and refer to the relevant literature and consult the relevant experts so as to build the index judgment matrix, also known as the pairwise comparison matrix, the pairwise comparison matrix is to express the comparison of the relative importance of all the factors in this layer against a certain factor (quasi-side or target) of the previous layer.

● Consistency Test:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \Leftrightarrow \begin{cases} a_{ij} > 0 \\ a_{11} = a_{22} = \cdots = a_{nn} = 1 \\ [a_{i1}, \dots, a_{in}] = k_i [a_{1n}, \dots, a_{1n}] \end{cases} \quad (7)$$

For a consistent array: then the normalized eigenvector $\{w_1, w_2, \dots, w_n\}$ corresponding to the largest eigenroot $\{w_1, w_2, \dots, w_n\}$ is taken, and $\sum_{i=1}^n w_i = 1$, w_i denote the weights of the i th factor in the lower layer on the degree of influence of a factor in the upper layer

Non-consistent array: the normalized eigenvector corresponding to its largest eigenroot is used as the weight vector $W = \{w_1, w_2, \dots, w_n\}$, then $AW = \lambda W$. The method of determining the weight vectors in this way is called the eigenroot method.

$$CI = \frac{\lambda - n}{n - 1} \quad (8)$$

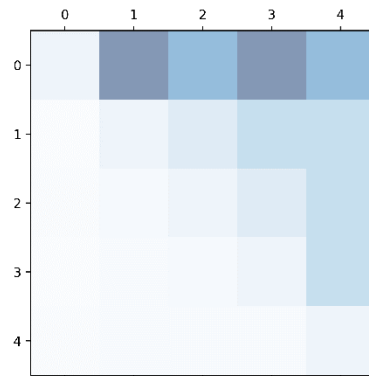
To determine whether the judgment matrix has consistency is mainly through the CI, RI and CR three consistency indicators to quantitative observation. If $CI=0$, then complete consistency; CI is close to 0, there is satisfactory consistency; the larger the CI, the more serious the inconsistency, and vice versa if the CI is smaller, the higher the consistency.

$$CR = \frac{CI}{RI} \quad (9)$$

RI is the corresponding average stochastic consistency index can be derived by looking up the table. CR is the consistency ratio, if $CR < 0.1$, then the consistency of the judgment matrix can be regarded as acceptable; otherwise, it is necessary to correct the judgment matrix.

● Judgment Matrix of the Five Factors at the Criterion Level:

A judgment matrix is constructed for the five factors at the criterion level, namely, historical value (HV), cultural significance (CS), economic contribution (EC), social impact (SI), and location risk (LR), and the weights of the matrix are based on the results of objective factual modeling based on the literature and the recommendations of the experts.



$$\begin{pmatrix} 1 & 7 & 5 & 7 & 5 \\ 0.143 & 1 & 2 & 3 & 3 \\ 0.200 & 0.500 & 1 & 2 & 3 \\ 0.143 & 0.333 & 0.500 & 1 & 3 \\ 0.200 & 0.333 & 0.333 & 0.333 & 1 \end{pmatrix} \quad (10)$$

The weight vector calculated by the eigenvalue method is $[0.2123, 0.164, 0.118, 0.081, 0.054]$; the CR value of the judgment matrix is 0.084. Because the CR of the judgment matrix is < 0.1 , it can be considered that the consistency of this judgment matrix is acceptable, and the test is passed. Similarly, the weights of the program layer and selection layer on the indicators can also be calculated using the above method.

6.4 Results of AHP Weighting

After deriving the individual weights of the criterion layer (C) on the Goal layer (G) and the weights of the indicators of the selection layer (A) on the criterion layer, the overall ranking of the selection layer can be derived by multiplying the $w_k^{(3)}$ matrix with the $w^{(2)}$ matrix.

By comparing the results of the weights of the selection layers, the ranking of the value of the building is derived i.e. to what extent we should protect the building.

Table 3: Selection Layer Overall Sort

$w^{(2)}$	0.583	0.164	0.118	0.081	0.054	Selection Layer Overall Sort
$w_k^{(3)}$	0.082	0.595	0.634	0.429	0.345	0.48869938
	0.236	0.276	0.192	0.429	0.109	0.26313172
	0.682	0.129	0.174	0.142	0.546	0.2481689
λ_k	3.001	3.006	3.009	3.000	3.054	--
CR_k	0.001	0.003	0.005	0.000	0.030	

Based on the results of the above table, we can find that the Statue of Liberty has the highest weight of 0.49, while Faneuil Hall and Fortress Monroe have relatively low weights of 0.26 and 0.25, respectively. Therefore, as a community leader, we should pay more attention to the preservation and protection of the Statue of Liberty.

7 A Letter to the Community

Dear Community members:

As you know, the Statue of Liberty, our beloved symbol of freedom and democracy, is under serious threat from the effects of climate change. Rising sea levels, storm surges, and increased storm intensity have already wreaked havoc on the Statue and its island, and could pose even greater risks in the future. The statue, a gift from the people of France to the people of the United States, has stood tall for more than a century, welcoming millions of visitors and immigrants to our shores. It is our duty and honor to preserve this iconic structure for ourselves and future generations.

Therefore, we launched a campaign to raise funds and awareness for the preservation and restoration of the Statue of Liberty. We consulted with experts from the National Park Service, the Statue of Liberty Ellis Island Foundation, and the French American Committee for the Restoration of the Statue of Liberty and developed a comprehensive plan to preserve and improve the Statue of Liberty and its island. Our plan includes the following measures:

Restoration: We will repair damage caused by weathering, corrosion and vandalism and restore the Statue to its original form and function. We will also replace the aging torch with a replica of Bartholdi's original design, gilded to his original plans. Based on the 1984-1986 restoration project, the restoration will cost approximately \$87 million.

Structural reinforcement: We will reinforce the internal structure of the statue, which consists of an iron skeleton and copper skin, to ensure its stability and safety. We will also reinforce the statue's foundation, as the statue is built on a historic fort that dates back to the War of 1812. Under the Infrastructure Improvement Program, which recently began in 2022, the structural reinforcement will cost approximately \$22 million.

Constructing disaster prevention facilities: we will build seawalls around the island to prevent flooding and erosion and install a drainage system to remove excess water. We will also upgrade electrical, sewer and fire protection systems to prevent power outages, pollution and fire hazards. Based on the estimated cost of similar projects in other coastal areas, the construction of the disaster prevention facilities will cost about \$50 million.

We believe that preserving the restoration of the Statue of Liberty is not only a matter of preserving a historic monument, but also a matter of preserving our values and identity as a nation. The Statue of Liberty represents our ideals of freedom, democracy and diversity, as well as our friendship with France and the world. It is a source of inspiration and hope for millions of people and a testament to our resilience and courage. We cannot afford to lose this precious asset or delay our action. We urge you to join us in this campaign and to spread the message to your family, friends and colleagues. Together, we can save the Statue of Liberty and ensure that her torch continues to shine for centuries to come.

Thank you for your interest and support.

Sincerely
Friends of the Statue of Liberty

8 Sensitivity Analysis

Sensitivity analysis determines the effect of a change in an independent variable on the outcome of a particular dependent variable under a given set of assumptions, and in this way we can test the robustness of the results. In sensitivity analysis, one or more parameters are varied within a reasonable range. At the same time, changes in the corresponding results are recorded and evaluated. In order to test whether the model results are sensitive to changes in the input parameters, we need to perform a sensitivity analysis.

The stability of the model is very important. It has to do with whether the model changes due to the interference of some other factors. In order to test the sensitivity of our model, we chose to perform a sensitivity analysis on a loss forecasting model that was built in predicting losses due to extreme weather impacts in the United States over the next 10 years. We increased and decreased the U.S. loss data for the past 10 years by 0.5% of the total losses, respectively, and input the changed data into the model we built. The prediction results for the three scenarios are shown below:

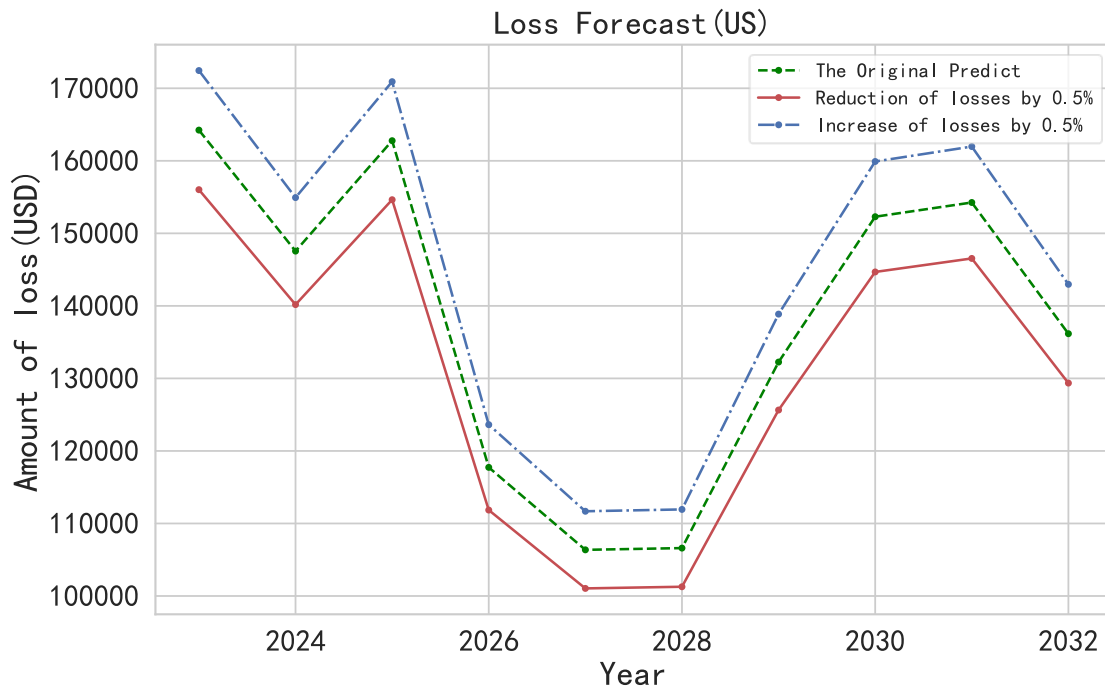


Figure 10: Predicting Loss Trends When Changing Past Loss Data by 0.5%

From the figure, we can see that when the loss data of the past ten years increases or decreases by 0.5%, the trend of its future loss data does not change, and the deviation is very small and within the acceptable range. Therefore, we can consider our model is stable and can solve the practical problems.

9 Model Evaluation and Further Discussion

9.1 Strengths

- **Insurance Model**

For question 1, this study selects countries with extreme weather events caused by the typical climates of North America and Asia as representatives, and creates an innovative (cost-benefit) insurance model using future projected losses from an economic perspective.

The base premium and loss equalization component of the insurance model can be changed depending on the frequency of extreme weather events in the region, which is applicable to a wide range of countries.

- **Hierarchical Analysis**

Systematic: The object is regarded as a system, according to the decomposition, comparison, judgment, synthesis way of thinking for decision-making. It becomes an important tool for system analysis developed after mechanistic analysis and statistical analysis;

Practicality: A combination of qualitative and quantitative, can deal with many practical problems that cannot be started with traditional optimization techniques, with a wide range of applications, and at the same time, this method enables decision makers and decision analysts to communicate with each other, and decision makers can even apply it directly, which increases the effectiveness of decision making;

Simplicity: Easy to calculate, the results are clear, with a medium level of education that people can understand the basic principles of hierarchical analysis and master the basic steps of the method, easy to be understood and mastered by decision makers. It is easy for decision makers to understand and grasp directly.

- **ARIMA Model**

It can deal with non-stationary time series and eliminate the trend and seasonal variation of the time series by means of differencing, so as to make it stable. Able to consider both the autoregressive part and the moving average part of the time series, capturing the historical trend and random fluctuations of the data, improving the accuracy and stability of forecasting.

9.2 Weaknesses

Limitations of Hierarchical Analysis

Confined to the old: Only one of the original solutions can be preferred, there is no way to arrive at a new and better solution;

Rough: The process of comparison, judgment, and calculation of results in this method is rough and not applicable to problems with higher precision;

Subjective: From building the hierarchical model to giving the pairwise comparison matrix, the human subjective factor has a great influence on the whole process, which makes it difficult for the results to be accepted by all decision makers. Of course taking the approach of expert group judgment is a way to overcome this shortcoming.

9.3 Further Discussion

- Improvement and extension of ARIMA model: firstly, adopting some improved methods, such as adopting nonlinear ARIMA model, generalized ARIMA model or fractal ARIMA model, to adapt to the characteristics of nonlinear time series; secondly, adopting some extended methods, such as adopting SARIMA model, STARIMA model or TARIMA model to deal with time series with seasonality, spatial correlation or threshold variation.

- Improvement and extension of AHP hierarchical analysis: first, some improvement methods are adopted, such as scaling only the lower or upper triangles of the judgment matrix, or adopting fuzzy hierarchical analysis or network hierarchical analysis in order to improve the consistency and reasonableness of the judgment matrix; second, some extension methods are adopted, such as combining grey theory, innovation theory, or intelligent algorithms, in order to generate new schemes or optimize existing schemes.

10 Conclusion

This paper develops an insurance model and a heritage conservation model and extends the application of the insurance model to assess the risk of site-specific construction. It not only provides a set of insurance pricing schemes for insurance companies to cope with future economic losses caused by extreme weather, but also provides quantitative support for the preservation of historic buildings in communities.

First, to predict economic losses due to extreme weather events, we used an autoregressive integral sliding average (ARIMA) model in representative areas on two different continents. Using this model, we predicted economic losses for the next 10 years in different regions and built an insurance model based on it. The insurance model is based on the financial data of Allstate, a well-known U.S. property insurance company, and through a break-even point analysis, the calculation verifies that the insurance company only has the probability of achieving the expected return if the number of local underwriters is more than 1291,522 accepting the property insurance price of 1479.65.

Secondly, based on the above model and with full consideration of the impact of extreme weather on real estate development and post-maintenance aspects, we selected the number of tornado occurrences, the number of hail occurrences, and the number of thunderstorm wind occurrences in all 50 U.S. states for a systematic clustering analysis to classify the U.S. states into high-risk, medium-risk, and low-risk areas, and combined with the principle of price discrimination, we adapted the insurance model to fit the risk level of the different regions.

Finally, in order to effectively address the protection of historic buildings in the community, we selected the building data of 26 communities across the U.S. with a certain degree of representativeness and built a comprehensive assessment model based on the weight assignment of the AHP model. The model quantitatively assesses the value of cultural artifacts from multiple dimensions, including historical value, cultural significance, and economic contribution, which will enable community leaders to more accurately identify important buildings and decide on relevant conservation measures on a quantitative basis.

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Appendices

Appendix 1
Introduce: US loss prediction python codes
<pre> %matplotlib inline %config inlineBackend.figure_format='retina' import pandas as pd import numpy as np import seaborn as sns sns.set(font='simhei',style='whitegrid',font_scale=1.6) import matplotlib.pyplot as plt import scipy.stats as stats plt.rcParams['font.sans-serif'] = ['SimHei'] plt.rcParams['axes.unicode_minus'] = False import warnings warnings.filterwarnings("ignore") from sklearn.ensemble import RandomForestRegressor from sklearn.model_selection import cross_val_score from statsmodels.tsa.stattools import adfuller from statsmodels.tsa.arima.model import ARIMA from sklearn.preprocessing import MinMaxScaler from statsmodels.graphics.tsaplots import plot_acf, plot_pacf from statsmodels.stats.diagnostic import acorr_ljungbox import itertools import warnings warnings.filterwarnings("ignore") df = pd.read_csv("events-US-1980-2021.csv", header=1) df['Name'] = [i.split('(')[0] for i in df['Name']] df['year'] = [int(str(i)[:4]) for i in df['Begin Date']] data = df.groupby('year')['Total CPI-Adjusted Cost (Millions of Dollars)'].sum() data_origin = data.copy() data1= { 'Year': range(1980, 2022), 'Value': [41058.8, 3133.3, 13661.6, 22902.8, 2893.1, 20264.8, 6049.1, 49009.7, 36270.6,36270.6,13129.4, 17626.6, 71870.4, 59109.2, 14893.9, 31966.6, 19689.7, 15422.1, 32801.9, 22046.0, 13978.0, 19575.2, 24458.6, 34533.4, 82880.4, 240564.5, 22678.0, 16939.2, 84622.3, 17826.2, 18133.4, 88181.3, 143234.9, 29046.1, 21901.2, 27797.6, 55055.3, 354704.7, 103344.7, 49928.1, 108673.1, 148035.6] } </pre>

```

df1= pd.DataFrame(data1)
df1['diff_1']=df1['Value'].diff(1)
df1['diff_1']=df1['diff_1'].fillna(0)
dataset=df1['diff_1']
data_oringin=dataset.copy()
result=adfuller(dataset)
scaler = MinMaxScaler()
scaled_data = pd.DataFrame(scaler.fit_transform(data.values.reshape(-1, 1)))
fig=plt.figure(figsize=(12,10))

p_min = 0
d_min = 0
q_min = 0
p_max = 4
d_max = 1
q_max = 4

results_bic=pd.DataFrame(index=['AR {}'.format(i) for i in range(p_min,p_max+1)],
                           columns=['MA {}'.format(i) for i in
range(q_min,q_max+1)])
for q,d,p in itertools.product(range(q_min,q_max+1),
                               range(d_min,d_max+1),
                               range(p_min,p_max+1)):
    if q==0 and p==0 and d==0:
        results_bic.loc['AR {}'.format(p),'MA {}'.format(q)]=np.nan
        continue
    try:
        model=ARIMA(scaled_data,order=(p,d,q))
        results=model.fit()
        results_bic.loc['AR {}'.format(p),'MA {}'.format(q)]=results.bic
    except:
        continue
df = pd.read_csv("events-US-1980-2021.csv", header=1)
df['Name'] = [i.split(' ')[0] for i in df['Name']]
df['year'] = [int(str(i)[:4]) for i in df['Begin Date']]
data = df.groupby('year')['Total CPI-Adjusted Cost (Millions of Dollars)'].sum()
data_origin = data.copy()
data1= {
    'Year': range(1980, 2022),
    'Value': [41058.8, 3133.3, 13661.6, 22902.8, 2893.1, 20264.8, 6049.1, 49009.7,
36270.6,36270.6,13129.4, 17626.6, 71870.4, 59109.2, 14893.9, 31966.6, 19689.7,
15422.1, 32801.9, 22046.0, 13978.0, 19575.2, 24458.6, 34533.4, 82880.4, 240564.5,

```

```

22678.0, 16939.2, 84622.3, 17826.2, 18133.4, 88181.3, 143234.9, 29046.1, 21901.2,
27797.6, 55055.3, 354704.7, 103344.7, 49928.1, 108673.1, 148035.6]
}
df1= pd.DataFrame(data1)
df1['diff_1']=df1['Value'].diff(1)
df1['diff_1']=df1['diff_1'].fillna(0)
dataset=df1['diff_1']
data_oringin=dataset.copy()
result=adfuller(dataset)
scaler = MinMaxScaler()
scaled_data = pd.DataFrame(scaler.fit_transform(data.values.reshape(-1, 1)))
fig=plt.figure(figsize=(12,10))

p_min = 0
d_min = 0
q_min = 0
p_max = 4
d_max = 1
q_max = 4

results_bic=pd.DataFrame(index=['AR{}'.format(i) for i in range(p_min,p_max+1)],
                           columns=['MA{}'.format(i) for i in
range(q_min,q_max+1)])
for q,d,p in itertools.product(range(q_min,q_max+1),
                               range(d_min,d_max+1),
                               range(p_min,p_max+1)):
    if q==0 and p==0 and d==0:
        results_bic.loc['AR{}'.format(p),'MA{}'.format(q)]=np.nan
        continue
    try:
        model=ARIMA(scaled_data,order=(p,d,q))
        results=model.fit()
        results_bic.loc['AR{}'.format(p),'MA{}'.format(q)]=results.bic
    except:
        continue

```

Appendix 2**Introduce: Risk Location Map python codes**

```
import pandas as pd
import numpy as np
import random
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

df=pd.read_csv("data.csv")
random_sequence = []
for _ in range(15625):
    random_sequence.append(random.randint(1, 2500))
random_sequence
df['all_risk']=random_sequence
fig = plt.figure(figsize=(10, 5))
ax = fig.add_subplot(111, projection='3d')

sc = ax.scatter(df['BEGIN_LON'], df['BEGIN_LAT'], df['all_risk'],
               c=df['all_risk'], cmap='inferno', marker='o', edgecolors='black')
cax = fig.add_axes([0.055, 0.15, 0.05, 0.7])
cbar = plt.colorbar(sc, cax=cax)
cbar.set_label('Risk Value')

ax.grid(True)
ax.view_init(elev=20, azim=-35)

ax.set_xlabel('Longitude')
ax.set_ylabel('Latitude')

plt.subplots_adjust(left=0.1, right=0.7, top=0.9, bottom=0.1)
```